

JAMIE WISE

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📍 Toronto, Ontario

EDUCATION

McGill University

Sept. 2022 – May 2026

Bachelor of Arts, Computer Science, Minor in Cognitive Science (GPA: 3.79/4.00)

Montreal, Quebec

- **Honours & Awards:** Graduated with Distinction, Social/Community Wellness Award at McGill CodeJam12.
- **Relevant Coursework:** Algorithms & Data Structures, Software Systems, Programming Languages, Algorithm Design, Applied Machine Learning, Reinforcement Learning, Computational Biology, Network Science.

University of New South Wales (UNSW)

Jan. 2025 – May 2025

Exchange Student, Computer Science

Sydney, Australia

- **Relevant Coursework:** Artificial Intelligence, Web Frontend Programming.

EXPERIENCE

McGill University – Department of Psychology

May 2024 – Aug. 2025

Research Assistant / Full-Stack Developer

Montreal, Quebec

- Collaborated with Professor Thomas Shultz to investigate GPT-4's discourse comprehension capabilities, turning open-ended research questions into testable tasks with clear prompts, controls, and evaluation criteria.
- Designed and implemented a full-stack AI-powered learning platform that leverages GPT's demonstrated capabilities to aid human comprehension, using Python (FastAPI), PostgreSQL, and React.
- Co-authored a peer-reviewed paper in *Royal Society Open Science* by synthesizing related work, analyzing experimental results, and communicating technical findings clearly and rigorously for publication.

SueApp

May 2023 – June 2023

Frontend Developer Intern

Tel Aviv, Israel

- Built a production-ready frontend in React and TypeScript integrated with a Node.js/Express backend for filing online legal claims, delivering a functional MVP within 5 weeks.
- Collaborated with a 4-person engineering team using Agile, Jira, and Git workflows to ensure smooth coordination and on-time delivery.
- Authored detailed onboarding documentation outlining Git practices, component structure, and local setup to streamline future developer onboarding.

PROJECTS

Clinical Converter 🗃️ 🔄 | Python, FastAPI, Next.js, Tailwind CSS, OpenAI API, CLI tooling

- Built a clinical data interoperability tool that converts legacy HL7 v2 hospital EMR messages into standardized FHIR R4 JSON outputs.
- Implemented a FastAPI (Python) backend to ingest, validate, and transform messages end-to-end; added a CLI and synthetic HL7 generators for testing.
- Developed an interactive Next.js web demo (via Vercel) with deterministic and LLM-generated summaries for quick human-readable views of the converted records.

Flash (AI Flashcard Generator) 🗃️ 🔄 | Python, JavaScript (React, Vite), FastAPI, Tailwind CSS, OpenAI API

- Built a flashcard generator that converts unstructured lecture text and PDFs into structured flashcards and summaries.
- Implemented a FastAPI service integrating the OpenAI API with custom prompting and an iterative concept-enumeration pipeline to extract key ideas while filtering noise.
- Designed a fuzzy alignment algorithm to match generated content back to the source text, reducing LLM token usage by ~88% while preserving accuracy, improving scalability and cost.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, Java, TypeScript, C, FastAPI, React, Next.js, PyTorch, NumPy/pandas.

Tools: Git, Linux/Bash, Jupyter, Vercel/Render.

PUBLICATIONS

Shultz, T. R., Wise, J. M., & Nobandegani, A. S. "Text understanding in GPT-4 versus humans."

Royal Society Open Science, 12, 241313. 2025. doi.org/10.1098/rsos.241313

- Peer-reviewed study comparing GPT-4's discourse comprehension to human performance.